

Waveguides consisting of single-crystal lithium niobate thin film and oxidized titanium stripe

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Abstract: Strip-loaded waveguides were fabricated by the direct oxidation of a titanium film based on the single-crystal lithium niobate. The method avoided the surface roughness problems that are normally introduced during dry etching of waveguide sidewalls. Propagation modes of the composite strip waveguide were analyzed by a full-vectorial finite difference method. The minimum dimensions of the propagation modes were calculated to be $0.7\ \mu\text{m}^2$ and $1.1\ \mu\text{m}^2$ for quasi-TM mode and quasi-TE mode at 1550 nm when the thickness of the LN layer and TiO_2 strip was 660 nm and 95 nm, respectively. The optical intensity was as high as 93% and was well confined in the LN layer for quasi-TM polarization. In this experiment, the propagation losses for the composite strip waveguide with 6 μm wide TiO_2 were 14 dB/cm for quasi-TM mode and 5.8 dB/cm for quasi-TE mode, respectively. The compact hybrid structures have the potential to be utilized for compact photonic integrated devices.

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OCIS codes: (160.3730) Lithium niobate; (160.2260) Ferroelectrics; (310.0310) Thin films.

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1. Introduction

A single-crystal lithium niobate (LN) thin film on an insulator (SiO₂) is called a LNOI and is an ideal platform for integrated photonics. It has attracted a lot of interest in recent years [1–3]. The good confinement and strong guiding of light due to the high-refractive-index contrast between LN and SiO₂, analogous to silicon on insulator (SOI), enables an ultra-compact photonic integrated circuit (PIC) [4]. Enhanced electro-optic and nonlinear optical effects are expected due to the high energy density in small regions [5]. Various ultra-compact photonic integrated devices such as electro-optical modulators [6], micro-ring resonator [7], micro-disk resonators [8,9], photonic crystals [10], and heterogeneous LN photonic devices [11] have been reported.

For photonic devices, the waveguide is the fundamental structure and the waveguides in LN thin film are typically made by the dry etching [12], proton-exchange [13], and dicing saw cutting [14]. However, the LN is relatively difficult to etch, and the rough dry-etched sidewall leads to a larger scattering loss. Proton-exchange in LNOI can only produce one polarized mode (quasi-TM mode) [13], and it is possible that the optical properties of LN decrease during the proton exchange process. Recently, a novel method that can realize much more compact LN devices on Si substrates has been introduced by adding an oxidized metal strip using selective oxidation of the refractory metal (SORM) on LN-on-Si [15]. It is estimated that the waveguide core size and bending radius can be reduced using this method [15]. Furthermore, the waveguide structures prepared by this method avoid the rough etched sidewalls by dry etching [16].

TiO₂ has attracted much attention as a prospective photonic material [17], and amorphous and polycrystalline TiO₂ thin film can be deposited at low temperatures (<650 K) using conventional methods [18]. The Ti indiffusion has become a standard fabrication method for optical waveguides in bulk LN material, and thus the deposition of Ti metal via photolithography and wet etching are quite mature. For Ti indiffusion, the diffuse temperature is around 1000 °C. However, for LNOI, the Ti indiffusion temperature cannot exceed 550 °C, because there will be some adhesion problem between SiO₂ and the LN substrate at high temperatures. Thus, it is interesting to explore the stripe-loaded waveguide on LNOI by oxidation of the Ti stripe. In this report, we employed the direct oxidation of the refractory metal (DORM), titanium (Ti), to form titanium dioxide (TiO₂) for strip-loaded waveguides on LNOI. This method was simple, and compatible with current mature technology of Ti-

indiffusion. The composition of the TiO_2 could be precisely controlled by the oxidation temperature, duration, atmosphere, etc. The refractive index of TiO_2 was about 2.3 at a 1.55 μm wavelength—similar to that of LN. The TiO_2 as a strip-loaded layer in combination with a LN layer could effectively form a composite strip waveguide. This avoided the difficulties of directly etching of LN. In the composite strip waveguides, the LN layer was the active region. The optimal structure could be designed to make most of the energy confined in the active region (LN layer). Therefore, the roughness of the TiO_2 sidewalls was expected to have little influence on the propagation loss versus the etched ridge waveguide [19]. More importantly, the hybrid structure could make use of the properties of different materials, leading to heterogeneous photonic devices.

In the paper, TiO_2 -LN composite planar and strip-loaded waveguides were studied by simulations and experiments. The single-mode condition as well as the power and the mode profile of the composite strip waveguide were simulated using a full-vectorial finite difference method [13,20]. Ellipsometry techniques were used to measure the refractive index and the film thickness of TiO_2 at 1550 nm. Composite strip waveguides with 4 μm , 5 μm , and 6 μm wide TiO_2 were fabricated and optically characterized. The mode size of a 6 μm wide composite strip waveguides was measured to be 4.3 μm^2 , and the propagation loss was 14 dB/cm for quasi-TM polarization. For quasi-TE polarization, the mode size was 4.5 μm^2 , and the propagation loss was 5.8 dB/cm. The realization of the composite strip waveguide could have a good prospect of fabricating more advanced and complicated integrated optical devices and circuits based on LNOI.

2. Simulation

Figure 1 showed the cross-section of the waveguide structure. The waveguide structure consisted of a TiO_2 strip loaded on the top of the LN layer, which was bonded to a SiO_2 layer on a LN substrate. This structure was called “strip-loaded waveguide” [21–24].

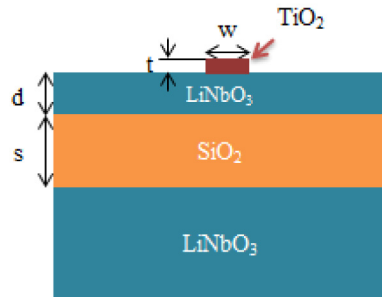


Fig. 1. Schematics of the waveguide structure cross-section. The structures from the bottom to the top are LN substrate, SiO_2 cladding, LN layer, and TiO_2 loading strip.

All of the LN films used for simulations and experiments were z-cut. The effective refractive index of the LN film was measured by a prism coupler (Model 2010 Metricon) at 1539 nm. The n_o and n_e of the LN film were 2.2119 and 2.1376, respectively. The birefringence was considered in the simulations. A full-vectorial finite difference method was used in the simulation. The strip-loaded waveguide generally supported two types of mode classified as E_{pq}^x (TE modes) and E_{pq}^y (TM modes), where the subscripts p and q indicate the number of extrema of the electric field in the x (horizontal) direction and y (vertical) direction, respectively. For the strip-loaded waveguide, the cut-off width of the strip was zero for the fundamental mode [13,21,25]. In our simulations, a zero cut-off width was always obtained. No matter how narrow the TiO_2 rib region was, there was always a waveguide mode. As shown in Fig. 2, the lines and symbols in blue and red were the single mode conditions for the TM and TE modes, respectively. The refractive index of TiO_2 is 2.3 at 1550 nm, which was measured by ellipsometry.

In the strip-loaded waveguide, the propagation properties could be controlled by changing the width w , the thickness t , the refractive index n of the loading TiO_2 strip and the thickness d of LN layer. For the planar waveguide, the single-mode condition in the y direction is $0 < d < 0.45 \mu\text{m}$ (TE) and $0.09 \mu\text{m} < d < 0.57 \mu\text{m}$ (TM), respectively, for a fixed TiO_2 layer thickness $t = 100 \text{ nm}$. Figure 2(a) showed the single mode condition in the x direction with the different thickness of the LN layer for a fixed TiO_2 strip thickness $t = 100 \text{ nm}$. The simulation results were consistent with the analysis by the Marcattili's method [26]. For example, for the TE mode, the single mode condition was $0 < w < 1.51 \mu\text{m}$ for a given LN layer thickness $d = 400 \text{ nm}$. For the TM mode, the single mode condition was $0 < w < 1.1 \mu\text{m}$ for a given LN layer thickness $d = 400 \text{ nm}$. However, in TM mode, the propagation loss of the fundamental mode was much larger when the thickness of the LN layer was less than 250 nm because in this situation, the TM mode was leaky because of $\text{TM} \leftrightarrow \text{TE}$ mode coupling at the sides of strip. In TE mode, it had a chance to become a truly guided mode [22–24]. Figure 2(b) showed the single mode condition in the x direction with the different thickness of the loading strip (TiO_2) for a fixed LN layer thickness $d = 400 \text{ nm}$. As the thickness of the TiO_2 loading strip increased, the single mode range decreased. For example, for the TE mode, the single mode condition was $0 < w < 1.66 \mu\text{m}$ for the 80 nm thick TiO_2 loading strip. For the TM mode, the single mode condition was $0 < w < 1.23 \mu\text{m}$ for the 80 nm thick TiO_2 loading strip.

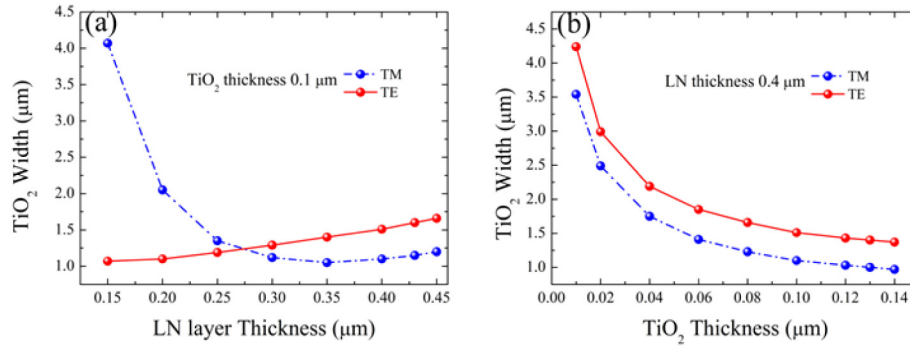


Fig. 2. (a) Single mode condition for the composite strip waveguide simulated at 1550 nm for different thickness values of the LN guiding layer; (b) Single mode condition for the composite strip waveguide simulated at 1550 nm for different thicknesses of the TiO_2 loading strip.

To be consistent with the experiments, the thickness of the LN layer was fixed at 660 nm in the simulation. To strengthen the nonlinear and electro-optical effects, a small mode size was needed. The simulation results of the relationship between the mode size and the width of the TiO_2 strip is shown in Fig. 3, there was a minimal value for the mode size at specific TiO_2 widths. For example, if the TiO_2 thickness was 95 nm , the minimum mode size was reached at $0.7 \mu\text{m}^2$ (product of the positions of $1/e$ light intensity in horizontal and vertical directions) when $w \sim 1 \mu\text{m}$ for the quasi-TM mode. It was $1.07 \mu\text{m}^2$ when $w \sim 1.3 \mu\text{m}$ for the quasi-TE mode. As the width of the TiO_2 strip increased from the minimum mode size position, the composite strip waveguide expanded, and thus the mode size became larger. When the width of the TiO_2 strip decreased, the confinement of light became weak. This also led to a larger mode size. The inset in Fig. 3 shows the intensity distributions of the minimum mode size when $t = 95 \text{ nm}$. We also simulated the mode size of the $\text{TiO}_2/\text{LiNbO}_3$ composite waveguide according to the structure of the one reported in [15]. Because the refractive index of TiO_2 was larger than that of Ta_2O_5 , the $\text{TiO}_2/\text{LiNbO}_3$ composite strip waveguide had a smaller mode size than that reported in [15] for TE mode. The small mode size implied strong light confinement—this benefited highly efficient nonlinear optical devices.

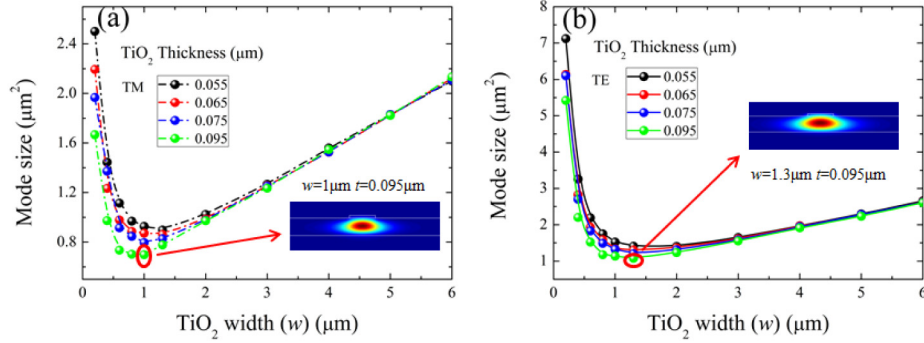


Fig. 3. (a) Relationship between the calculated mode size and width of the TiO_2 strip for quasi-TM mode; inset: a mode size as small as $0.7 \mu\text{m}^2$ was obtained with a $1 \mu\text{m}$ wide TiO_2 strip; (b) Relationship between the calculated mode size and width of the TiO_2 strip for quasi-TE mode; inset: a mode size as small as $1.07 \mu\text{m}^2$ with a $1.3 \mu\text{m}$ wide TiO_2 strip.

For the composite strip waveguide, the optical power was mainly divided into two parts—one part was in the LN layer and the other was in the TiO_2 loading strip. This required most of the optical power concentrated in the LN layer especially in nonlinear optics and E-O applications. Figure 4 showed the optical powers in the whole LN layer and the TiO_2 strip region using the TiO_2 width as the parameter. The left and right vertical axis indicated optical powers in the LN layer and the TiO_2 strip, respectively. As shown in Fig. 4(a), when the TiO_2 strip thickness was less than 100 nm, the optical power in the LN layer was more than 93% for quasi-TM mode. For quasi-TE mode, when the TiO_2 thin film thickness was less than 100 nm, the optical power confined in the LN layer was more than 89% (see Fig. 4(b)). The yellow line in Figs. 4(a) and 4(b) indicate that the optical power in the $2 \mu\text{m}$ wide TiO_2 strip increased as a function of TiO_2 strip thickness. For TM mode, the light energy in TiO_2 changed from nearly 0 to about 3.7% when the TiO_2 thickness changed from 15 nm to 135 nm. For TE mode, the light energy in TiO_2 changed from 0.1% to about 7.4% when the TiO_2 thickness was 15 nm to 135 nm. Figure 4(a) shows that the optical power in the LN layer would slightly decrease with shrinking TiO_2 width for quasi-TM mode when the thickness of the TiO_2 strip was fixed. Figure 4(b) showed that the optical power in the LN layer would slightly increase with shrinking TiO_2 width for quasi-TE mode when the thickness of the TiO_2 strip was fixed. Therefore, the thickness and width of the TiO_2 strip had some influence on the optical power distribution in the LN layer and the TiO_2 stripe.

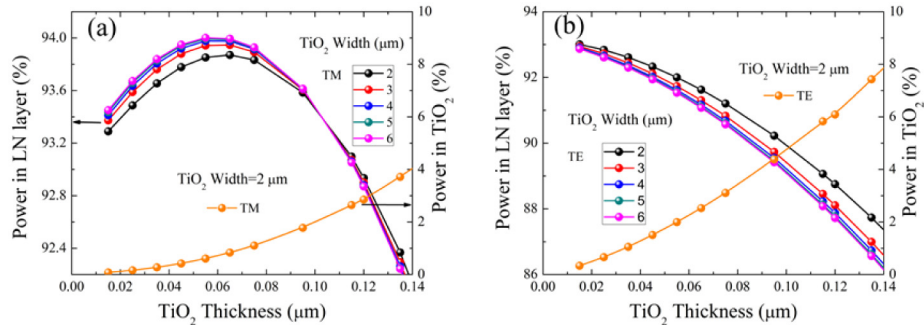


Fig. 4. (a) Relationship between optical power in the LiNbO_3 layer and the thickness and the width of the TiO_2 strip as well as the relationship between optical power in the TiO_2 strip and the thickness of the TiO_2 strip for quasi-TM mode; (b) Relationship between optical powers in LiNbO_3 layer and the thickness and the width of TiO_2 strip as well as the relationship between optical powers in the TiO_2 strip and the thickness of the TiO_2 strip for quasi-TE mode.

3. Experiment and results

Four identical z-cut LNOI samples (#1, #2, #3, and #4) were prepared by crystal ion slicing and wafer bonding technology based on a previously described process [1]. Sample #1 was used to fabricate the composite strip waveguides. Sample #2 was used to analyze the concentration of Ti as a function of depth, and Sample #3 was used to investigate the refractive index of the TiO_2 thin film.

The fabrication procedure for sample #1 is shown in Fig. 5. First, a titanium film with a thickness of 81 nm was deposited on the surface of the LN layer via RF magnetic sputtering. The thickness of the Ti film was measured with a step profiler. Next, a positive photoresist mask with 5 μm and 6 μm wide strips was crated via photolithography. Then, the exposed Ti was wet etched with Ti etchant ($\text{H}_2\text{O} + \text{H}_2\text{O}_2 + \text{HF}$), and the photoresist was removed by acetone. The remaining Ti was oxidized in a furnace with oxygen flow at 500 $^\circ\text{C}$ for 15 h. After oxidation, a composite strip waveguide consisting of a TiO_2 strip on the LN layer, was formed. Finally, the two facets of the waveguides were carefully polished to facilitate the end face coupling. The length of the waveguide was about 2.4 mm. The 4 μm composite strip waveguides were fabricated on sample #4 using a similar process as sample #1 except that the Ti thickness was 67 nm. Figure 6(a) presents scanning electron microscopy (SEM) data of the cross-section of the fabricated waveguide. Figure 6(b) shows the top view of the composite strip waveguide by optical microscopy.

To explore whether Ti diffused in LN or not, a much thicker Ti layer than sample #1 was prepared on sample #2 under the same condition as sample #1. After the Ti layer was oxidized in a furnace with an oxygen flow at 500 $^\circ\text{C}$ for 15 h. The concentration of Ti was analyzed by Secondary Ion Mass Spectroscopy (SIMS). Figure 6(c) showed the Ti concentration profile as a function of depth. This showed that Ti did not diffuse significantly in z-cut LN thin films at 500 $^\circ\text{C}$. The thickness of the TiO_2 film was about 190 nm as shown in Fig. 6(c). This was consistent with the value measured by the step profiler. To investigate the refractive index of the TiO_2 layer, Ti layer with the same thickness as sample #1 was prepared on sample #3 using the same RF magnetic sputtering experimental conditions. After the Ti layer was oxidized in a furnace with oxygen flow at 500 $^\circ\text{C}$ for 15 h, the refractive index and the film thickness of TiO_2 were measured to be around 2.3 and 95 nm, respectively using ellipsometry at 1550 nm (JobinYvon UVISSEL 2). The thickness of the TiO_2 layer was consistent with the value measured by the step profiler.

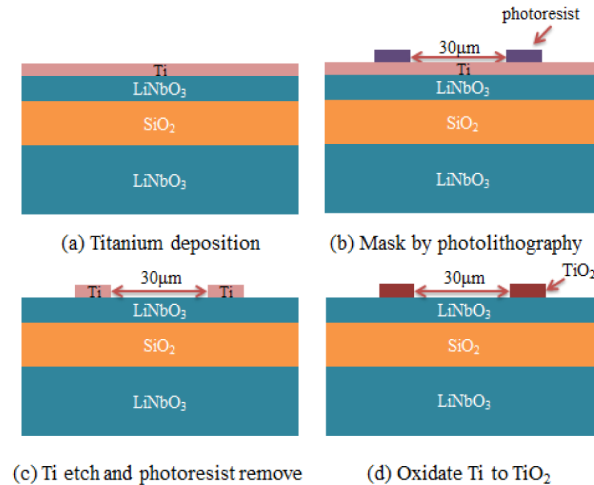


Fig. 5. The process steps of directed oxidation of titanium to form LN strip waveguide.

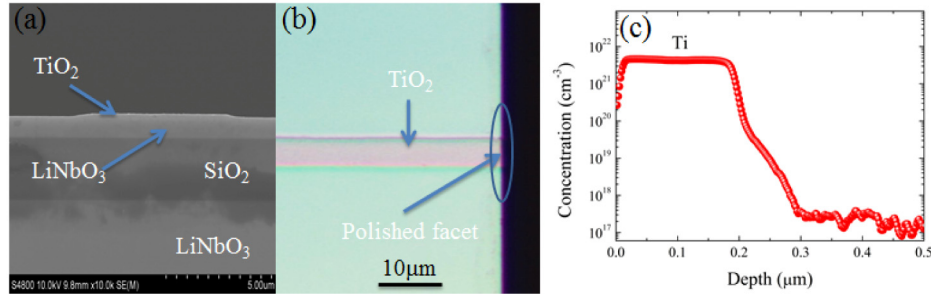


Fig. 6. (a) SEM image of a cross-section of a fabricated waveguide via the DORM method; (b) Optical microscope image (top view) of a 6 μm wide TiO_2 strip loaded on top of the LN layer; (c) Ti concentration as a function of depth obtained by SIMS in a TiO_2 /LNOI sample annealed at 500 $^\circ\text{C}$ for 15 h in dry O_2 atmosphere. The TiO_2 film was about 190 nm thick.

A tunable semiconductor laser (Santec TSL-210) was used as the near-infrared light source (1260 nm–1630 nm). The linear polarized light emitted from this laser was transmitted through a polarization-maintaining fiber and then rotated to the TM or TE polarized direction by a rotator. The light was coupled into the waveguide by the lensed tip of the fiber and the output light was collected by a $40\times/0.65$ objectives. A knife-edge based beam profiler (BM-7, Coherent Inc.) and germanium photodiode were used to record the near-field intensity profile and the power of the incident light, respectively. The BM-7 utilized 7 knife-edges to scan across the beam and capture the intensity distribution of the light. The larger number of knife-edges (7 in our case) created better details on the non-Gaussian beam and reconstructed a profile that matched the real beam. Figure 7 showed the measured mode distribution (quasi-TM polarization and quasi-TE polarization) and the corresponding simulated mode distributions, respectively. The measured and theoretical mode sizes were defined as the product of the $1/e$ intensity in horizontal and vertical direction of the mode profile. For quasi-TM polarization, the measured mode size was about $4.3\ \mu\text{m}^2$, and the theoretical value was $2.1\ \mu\text{m}^2$. For quasi-TE polarization, the measured mode size was about $4.5\ \mu\text{m}^2$, and the theoretical value was $2.5\ \mu\text{m}^2$. The resolution of the objective ($\sim 1.5\ \mu\text{m}$) is larger than the vertical dimension of the composite strip waveguide ($0.6\ \mu\text{m}$)—this leads to larger mode sizes in the measurement.

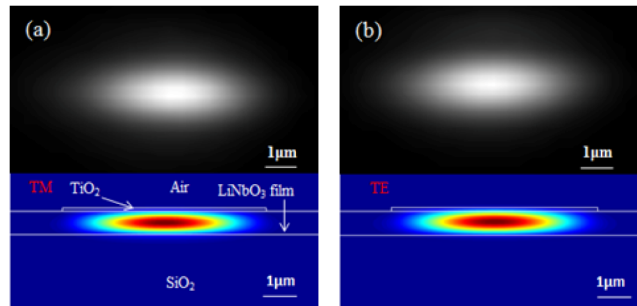


Fig. 7. Measured and simulated near-field intensity distributions of the fundamental quasi-TM mode (a) and quasi-TE mode (b) guided in the 6 μm wide composite strip waveguides at 1.55 μm . Mode sizes of the $4.3\ \mu\text{m}^2$ (quasi-TM mode) and $4.5\ \mu\text{m}^2$ (quasi-TE mode) were obtained by measurement and mode sizes of $2.1\ \mu\text{m}^2$ and $2.5\ \mu\text{m}^2$ were obtained by simulation.

The propagation loss α of the composite strip waveguide was evaluated by the Fabry-Perot resonator method. The reflectivity R of the end-faces was obtained via a finite difference time domain (FDTD) simulation. The mode propagation loss α was determined by analyzing the contrast K of the cavity resonances [27]. The loss α and the related parameters were defined by the following equations:

$$\alpha = \frac{4.34}{L} (\ln R - \ln \tilde{R}) \quad \text{with} \quad R = \frac{1}{K} (1 - \sqrt{1 - K^2}) \quad \text{and} \quad K = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} \quad (1)$$

Figure 8 showed the measured transmission spectrum of quasi-TE mode and quasi-TM mode. The $I_{\max}$ and $I_{\min}$ were the maximum and minimum intensity of the transmitted light, which were extracted from Fig. 8. The modal reflectivity R of the 6 μm wide composite strip waveguide end face were 0.105 and 0.216 for quasi-TM mode and quasi-TE mode as simulated by the FDTD method, respectively [22]. For quasi-TM mode, the loss α was 14.0 dB/cm. For quasi-TE mode, the loss α was 5.8 dB/cm. The TiO_2 is prone to polycrystalline forms whose grain boundaries could lead to scattering. Scattering of light in the TiO_2 strip region could cause a high loss as reported in the TiO_2 waveguides on SiO_2 [18,22]. Another possible explanation was that a small fraction of the higher order modes was probably excited by end-face coupling for multi-mode waveguide despite the fundamental mode being on display in Fig. 8. This resulted in an error for loss estimation [28]. The propagation losses of 4 μm and 5 μm wide waveguides were measured to be about 7.6 (17.2), and 9.8 (15) dB/cm for quasi-TE (TM) mode, respectively. Further investigation on the source of the loss discrepancy among the waveguides with different widths is underway. Versus the literature [15], the propagation losses of the $\text{TiO}_2/\text{LiNbO}_3$ composite waveguide in this paper had approximately the same value.

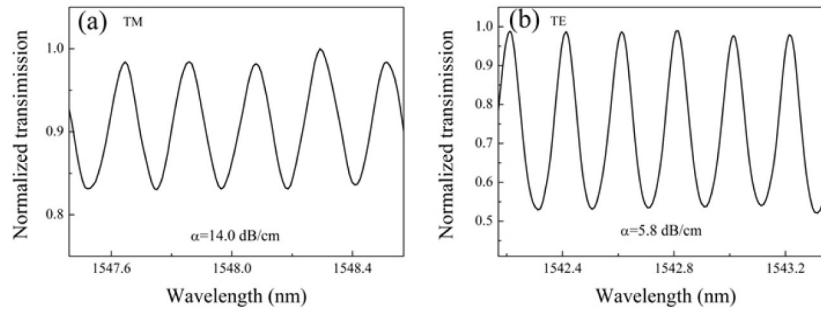


Fig. 8. (a) Normalized transmission of quasi-TM polarized light in the 6 μm composite strip waveguides as a function of wavelength; (b) Normalized transmission of quasi-TE polarized light in the 6 μm composite strip waveguides as a function of wavelength.

4. Conclusions

In conclusion, this work demonstrated the formation of stripe-loaded waveguides by direct oxidation of titanium stripes on the single-crystal lithium niobate thin film. Propagation modes of the stripe waveguide are discussed in detail at 1550 nm by simulation and direct measurement. The quasi-TM and TE mode were calculated to be 0.7 μm^2 and 1.1 μm^2 , respectively. This was much smaller than the conventional Ti-diffused LN waveguide (around 10 μm^2). In addition, more than 93% (89%) of the optical intensity was well confined in the LN layer for quasi-TM (TE) mode in the LN layer. The thickness of the TiO_2 layer was less than 100 nm. According to experiments, the mode size of a composite strip waveguide of 6 μm width was 4.3 μm^2 for quasi-TM mode and 4.5 μm^2 for quasi-TE mode. The propagation losses were evaluated to be 14 dB/cm for quasi-TM mode and 5.8 dB/cm for quasi-TE mode, respectively. The simulation and successful fabrication of composite strip LN waveguides in LNOI with TiO_2 as the loaded material will improve future research efforts on high-performance and complicated nonlinear and E-O integrated photonic devices and circuits.

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